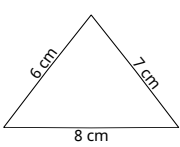




9A. Mensuration

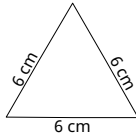
1. Find the perimeter of each of the following figures.



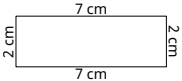
(a)



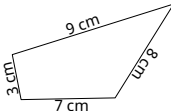
(b)



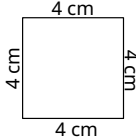
(c)



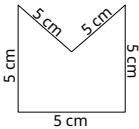
(d)



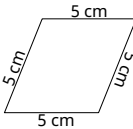
(e)



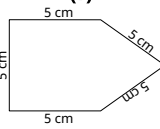
(f)



(g)

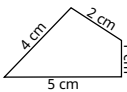


(h)

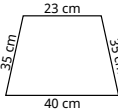


(i)

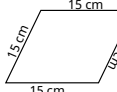
2. Find the perimeter of each of the following figures :



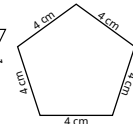
(a)



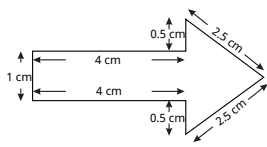
(b)



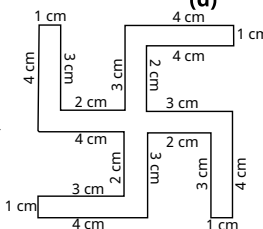
(c)



(d)



(e)



(f)

3. Find the perimeter of the ΔABC if

- (a) $AB = 7$ cm, $BC = 8$ cm and $CA = 9$ cm
- (b) $AB = 12$ cm, $BC = 5$ cm and $CA = 13$ cm
- (c) $AB = 4$ m, $BC = 3$ m and $CA = 6$ m
- (d) $BC = 1$ m 75 cm, $CA = 2$ m 50 cm and $AB = 3$ m

4. Find the perimeter of the equilateral triangle in which each side is: (a) 4 cm (b) 5 m (c) 3 m 25 cm

5. Find the perimeter of the rectangle whose

- (a) length = 13 cm, breadth = 5 cm
- (b) breadth = 7 m, length = 8 m
- (c) length = 17 cm, breadth = 15 cm
- (d) breadth = 22 cm, length = 25 cm
- (e) length 16.8 cm, breadth 6.2 cm

(f) length = 2 m 25 cm, breadth 1 m 50 cm

(g) length = 8 m 5 dm, breadth = 6 m 8 dm

6. Find the perimeter of the square in which each side is: (a) 7 cm (b) 5 m (c) 12 m 50 cm

7. Find the length of the boundary wall of a rectangular garden which is 132 m long and 80 m broad.

8. Find the perimeter of a rectangular flower-bed which is 7 m 10 cm long and 3 m 20 cm broad.

9. The side of a square field is 160 m. If a farmer walks around its boundary once, find the total distance he has to cover.

10. The field shown in the figure is to be fenced by 3 wires going around the field together. Find the total length of the Wire required for fencing. If the wire costs Rs 10 per metre, find the total cost offencing.

11. A path is to be made around a park whose shape is that of an equilateral triangle. Each side of the park is 25 m long. Find the cost of making the path if it costs Rs 50 per metre to make the path.

12. Find the perimeter of

(a) a triangle of sides 7.8 cm, 6.5 cm and 5.9 cm,

(b) an equilateral triangle of side 9.4 cm,

(c) an isosceles triangle with equal sides 8.5 cm each and third side 7 cm.

13. Find the perimeter of

(a) a regular pentagon of side 8 cm,

(b) a regular octagon of side 4.5 cm,

(c) a regular decagon of side 3.6 cm.

14. Find the circumference of a circle whose radius is

(a) 28 cm

(b) 10.5 cm

(c) 3.5 m

15. Find the circumference of a circle whose diameter is

(a) 14 cm

(b) 35 cm

(c) 10.5 m

16. Find the area of the circles with the following radius: (Take $\pi = \frac{22}{7}$)

(a) 14 cm

(b) 28 mm

(c) 21 cm

17. Find the radius of a circle whose circumference is 176 cm.

18. Find the diameter of a wheel whose circumference is 264 cm.

19. Find the distance covered by the wheel of a car in 500 revolutions if the diameter of the wheel is 77 cm.

20. The diameter of the wheel of a car is 70 cm. How many revolutions will it make to 1.65 km?

21. Find the area of each of the following rectangles:

(a) Length = 15 m and breadth = 12 m

(b) Length = 21 m and breadth = 14 m

(c) Length = 150 cm and breadth = 85 cm

(d) Length = 3 m 25 cm and breadth = 2 m

(e) Length = 8 km and breadth = 2 km 500 m

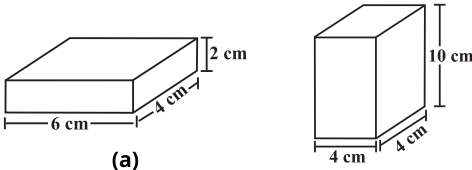
22. Find the length of the rectangle whose

(a) area = 275 sq m and breadth = 23 m

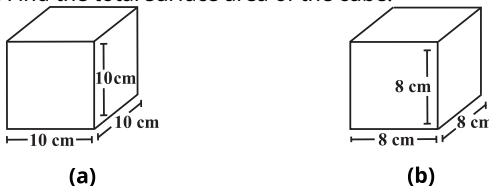
- (b) area = 300 sq m and breadth = 15 m
(c) area = 512 sq cm and breadth = 16 cm
23. Find the breadth of the rectangle whose
(a) area = 12750 sq m and length = 150 m
(b) area = 5500 sq m and length = 110 m
(c) area = 1926 sq cm and length = 107 cm
24. Find the area of the square whose each side is:
(a) 15 m (b) 18 m (c) 20 m
(d) 3 m 40 cm (e) 5 m 50 cm
25. A garden is 800 cm long and 300 cm broad. Find its length and breadth in metres and area in sq m.
26. Find the area of a rectangular flower-bed in sq cm, if its length is 7 m 30 cm and breadth is 4 m 50 cm.
27. Find the area of the floor of a square room in square metres whose each side is 800 cm.
28. How many bricks will be required to lay a path 120 m long and 2.4 m broad if a brick is 24 cm long and 15 cm wide?
29. A field is 140 m long and 36 m broad. Labourers are engaged to plough the field. If a labourer can plough 120 sq m a day, how many labourers are to be engaged to plough the field in a day?
30. Find the cost of tiling a courtyard 30 m long and 15 m broad at the rate of ₹ 5 per sq cm.
31. Find the cost of painting the walls of a room if the room is 4 m 50 cm high and each side of the room is 6 m long. The cost of painting is Rs 10 per sq m.

9B. Volume and Surface Area of Solids

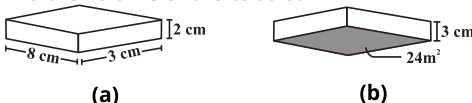
1. Find the total surface area of the cuboid:



2. Find the total surface area of the cube:



3. Find the volume of the cuboid:



4. Find the volumes of the following cuboids and cubes by counting the unit cubes of volume 1 cu cm.

5. Find the volume of the cube whose each edge is:
(a) 5 cm (b) 4 m (c) 6 cm (d) 7 m

6. Find the volume, lateral surface area and the total surface area of a cube each of whose edges measures: (a) 7 m (b) 5.6 cm (c) 8 dm 5 cm

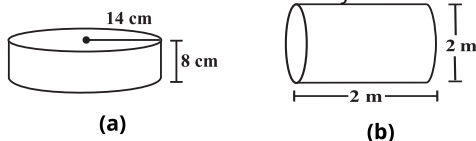
7. Find the volume of the cuboid whose dimensions are:

- (a) length = 5 m, breadth = 4 m, height = 3 m
(b) length = 12 m, breadth = 5 m, height = 4 m
(c) length = 48 cm, breadth = 36 cm, height = 24 cm

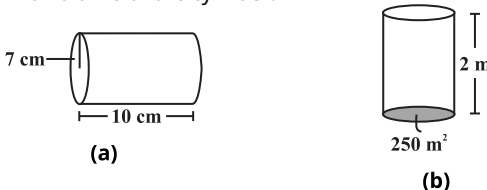
8. Find the volume, lateral surface area and the total surface area of the cuboid whose dimensions are:

- (a) length = 22 cm, breadth = 12 cm and height = 7.5 cm
(b) length = 15 m, breadth = 6 m and height = 9 dm
(c) length = 24 m, breadth = 25 cm and height = 6 m
(d) length = 48 cm, breadth = 6 dm and height = 1 m

9. Find the total surface area of the cylinder:



10. Find volume of the cylinder:



11. Find the volume, curved surface area and total surface area of each of the cylinders whose dimensions are:

- (a) radius of the base = 7 cm and height = 50 cm
(b) radius of the base = 5.6 m and height = 1.25 m
(c) radius of the base = 14 dm and height = 15 m

12. Find the volume of a cube whose volume is three times the volume of a cuboid of dimensions 15 cm $\times$ 12 cm $\times$ 10 cm.

13. Find the volume of a cuboid whose volume is five times the volume of a cube whose each edge is 10 m.

14. Show that the volume of a cube, whose each edge measures 12 m, is 9 times the volume of a cuboid of dimensions 8 m $\times$ 6 m $\times$ 4 m.

15. A cuboid measures 48 m by 32 m by 24 m and each edge of a cube measures 16 m. Find the ratio of their volumes.

16. Find the volume of oil that can be poured into a cuboidal tin whose internal dimensions are 8 cm $\times$ 4 cm $\times$ 10 cm.

17. Find the volume of air in a room 10 m long, 6.5 m wide and 5 m high.

18. How many bricks each 25 cm long, 10 cm wide and 7.5 cm thick, will be required for a wall 15 m long, 2.5 m high and 1.25 m thick?

19. How many bricks, each of dimensions 18 cm by 12 cm by 10 cm, will be required to build a wall 12 m long, 80 cm thick and 3.6 m high? Find the total cost of bricks at Rs 750 per 1000.